Chapter 13, Problem 39P

Problem

A 1,200/240-V rms transformer has impedance $60\angle -30^\circ\ \Omega$ on the high-voltage side. If the transformer is connected to a $0.8\angle 10^\circ$-Ω load on the low-voltage side, determine the primary and secondary currents when the transformer is connected to 1,200 V rms.

Step-by-step solution

Step 1 of 4

The primary voltage of the transformer is, $\mathbf{V}_1 = 1200\text{ V}$

The secondary voltage of the transformer is, $\mathbf{V}_2 = 240\text{ V}$

Determine the turns ratio or transformation ratio of the transformer.

$$\begin{aligned} n &= \frac{N_2}{N_1} \\ &= \frac{\mathbf{V}_2}{\mathbf{V}_1} \\ &= \frac{240}{1200} \\ &= 0.2 \end{aligned}$$

Step 2 of 4

The value of load impedance is, $\mathbf{Z}_L = 0.8\angle 10^\circ\ \Omega$

Determine the load impedance referred to the primary side.

$$\begin{aligned} \mathbf{Z}_R &= \frac{\mathbf{Z}_L}{n^2} \\ &= \frac{0.8\angle 10^\circ}{(0.2)^2} \\ &= \frac{0.8\angle 10^\circ}{0.04} \\ &= 20\angle 10^\circ\ \Omega \end{aligned}$$

Step 3 of 4

The impedance in the primary side is, $\mathbf{Z}_P = 60\angle -30^\circ\ \Omega$

Determine the input impedance of the transformer.

$$\begin{aligned}\mathbf{Z}_{\text{in}} &= \mathbf{Z}_P + \mathbf{Z}_R \\ &= 60\angle -30^\circ + 20\angle 10^\circ \\ &= 76.41\angle -20.31^\circ\ \Omega\end{aligned}$$

Determine the primary current, $\mathbf{I}_1$.

$$\begin{aligned}\mathbf{I}_1 &= \frac{\mathbf{V}_1}{\mathbf{Z}_{\text{in}}} \\ &= \frac{1200}{76.41\angle -20.31^\circ} \\ &= 15.7\angle 20.31^\circ\ \text{A}\end{aligned}$$

Therefore, the value of primary current in the transformer, $\mathbf{I}_1$ is $\boxed{15.7\angle 20.31^\circ\ \text{A}}$.

Step 4 of 4

Determine the secondary current, $\mathbf{I}_2$.

$$\begin{aligned}\mathbf{I}_2 &= \frac{\mathbf{I}_1}{n} \\ &= \frac{15.7\angle 20.31^\circ}{0.2} \\ &= 78.5\angle 20.31^\circ\ \text{A}\end{aligned}$$

Therefore, the value of secondary current in the transformer, $\mathbf{I}_2$ is $\boxed{78.5\angle 20.31^\circ\ \text{A}}$.